D J Tharun

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🚀 PROFILE

Class 12 Science student (PCM) and Senior Head Boy specializing in the intersection of **AI, Quantum Computing, and Civic Innovation**. Proven track record of transitioning complex technical projects from ideation to national-level recognition. Passionate about leveraging **agentic workflows** and **IoT** to build resilient, community-centric systems.

🛠️ TECHNICAL PROJECTS

Symbiotic City | National Level Finalist (ATL Competition) * Developed a modular urban infrastructure prototype aligned with **UN SDG 11** (Sustainable Cities).

- Integrated **Raspberry pi, Arduino** and **NodeMCU** with **MQTT protocols** for real-time resource monitoring and data visualization.
- Architected a self-sustaining data loop using **energy-harvesting sensors** to minimize external power reliance.

School EVM v3.0 (Digital Democracy) * Designed and deployed a **fingerprint-authenticated** electronic voting machine for school-wide elections.

- Eliminated paper waste and increased tallying efficiency by **100%** using **C++** and **ESP32** biometric integration.
- Managed a centralized monitoring system ensuring secure, transparent voting for a student body of 1,000+.

FYTO (Smart Agriculture IoT) * Engineered an automated plant care system utilizing **Raspberry Pi** and a custom array of environmental sensors.

- Integrated **OpenWeather API** to automate irrigation logic based on hyper-local weather forecasting.

🎽 LEADERSHIP & IMPACT

Senior Head Boy | PBKV *(2025 – 2026)* * Elected representative for 1,000+ students, serving as the strategic liaison between the student body and school administration.

- Spearheaded community outreach initiatives in collaboration with local law enforcement for anti-addiction awareness campaigns.

Social Advocate & Performer * Lead performer in a high-visibility theatrical production focused on **Transgender Rights**, designed to drive empathy and policy dialogue within the academic community.

🎯 SKILLS & COMPETENCIES

- **Hardware/IoT:** Arduino, NodeMCU, ESP32, Raspberry Pi, Biometric & Environmental Sensors.
- **Software:** Python, C++, HTML, Agentic IDEs (Google Antigravity), AI-Assisted Development.
- **Interdisciplinary:** Quantum Computing fundamentals, Sustainable Urban Design (SDG 11).
- **Soft Skills:** Public Speaking, Crisis Management, Strategic Planning, Cross-functional Leadership.

🏹 THE "HUMAN" SIDE (Beyond the Lab)

- **Archery:** Developed "The Archer's Focus"—applying extreme mental discipline and precision to engineering challenges.
- **Culinary Art:** Self-taught "Kitchen Chemist" specializing in Burmese cuisine; applying the same rigor to recipe optimization as debugging code.
- **Cricket:** Competitive All-rounder; focused on team strategy and calm decision-making under high-pressure match conditions.

📜 EDUCATION

Class 12 (Science - PCM) | PBKV *Expected Completion: March 2026 (CBSE Board)*

Verified Note: While many school-based leadership roles were hands-on and informal, contact details for faculty mentors and project supervisors are available as references upon request.